

Unrelated Is Not Opposed: A Signed Scale for Meaning Preservation

Anonymous Submission

Abstract

Metrics for meaning preservation report how much meaning two sentences share, on a scale from 0 to 100. That scale cannot express opposition. Because contradicting sentences are about the same thing, they overlap heavily, and a released model scores “I drink milk” against “I do not drink milk” at 58 out of 100. We propose a signed scale on $[-100, 100]$, obtained by multiplying a magnitude head by $1 - \alpha p_{\text{contra}}$ from a three-way polarity head. The form is a product so that unrelated sentences score 0 whatever their polarity. Thresholded token overlap ranks polarity better than the released metric does, 63.5% AUC against 60.8%. Across 140 runs a polarity head reaches 91.1% macro- F_1 and 98.7% AUC, derived-polarity augmentation leaves task performance unchanged while raising generated sanity suites from 79–90% to 98–99% on all seven encoders, and composition puts 84% of SICK contradictions below zero where the magnitude alone puts none.

1 Introduction

Assessing whether a rewritten sentence preserves the meaning of its source is a prerequisite for evaluating text simplification, summarisation and translation (Saggion, 2017). The metrics built for it, whether embedding-based (Zhang et al., 2020; Reimers and Gurevych, 2019), question-based (Scialom et al., 2021) or trained end to end (Maddela et al., 2023; Beauchemin et al., 2023), all return a single non-negative number, where 0 means the sentences have nothing in common and 100 means the meaning is intact.

That output shape leaves no room for opposition. Two sentences that contradict each other are, by construction, about the same thing, so they share most of their lexical material and most of their subject matter, and a magnitude metric reads that overlap as preserved meaning. A released model scores “I drink milk” against “I do not drink milk” at 58.3, and the SICK contradiction “two dogs are wrestling and hugging” against “there is no dog wrestling and hugging” at 51.7. Both land on the positive half of a scale whose positive half

means the meaning survived. Baral (2026) quantified the same effect on legal text, where minimal perturbations that reverse legal force while preserving most tokens move embedding metrics over a small fraction of their range. These metrics are not imprecise on such pairs; they are confidently wrong in the direction that matters.

The failure is measurable on a released metric. Evaluated on a held-out set of 12 000 class-balanced pairs, the published MeaningBERT model ranks agreement above contradiction with an AUC of 60.8%, and on SICK (Marelli et al., 2014) alone it reaches 48.0%, scoring contradictions marginally higher than agreements. A thresholded count of shared tokens reaches 63.5% on the same split. A metric trained on human meaning-preservation judgements therefore orders polarity no better than lexical overlap.

This paper proposes a signed scale on $[-100, 100]$, where -100 denotes opposed meaning and 0 denotes no relation. Our contributions are:

1. **A signed scale obtained by composition** (Section 3). No corpus carries human-annotated signed scores, so learning the scale end to end would fit a model to our own conversion rule. We instead compose two heads whose targets both have abundant human labels, and calibrate the composition on the single corpus annotating both dimensions over the same pairs.
2. **A 140-run grid** (Section 5, Section 6). Seven encoders, ten seeds and two corpus conditions, on a class- and corpus-stratified split with source-sentence leakage removed in both directions.
3. **Three measured results** (Section 6): a polarity head reaches 91.1% macro- F_1 over ten seeds and, on a single run, ranks entailment above contradiction at 98.7% AUC; derived-polarity augmentation leaves task performance unchanged while repairing behaviour at the ends of the scale on all seven encoders; and inference pretraining buys reasoning under negation rather than aggregate accuracy at large size.

2 Related Work

Meaning preservation metrics. BERTScore (Zhang et al., 2020) computes token-level similarity from contextual embeddings, SARI (Xu et al., 2016) scores edit operations, QuestEval (Scialom et al., 2021) asks questions of both texts, and LENS (Maddela et al., 2023) and MeaningBERT (Beauchemin et al., 2023) are trained against human judgements. All emit a magnitude, and all are evaluated by correlation with human ratings plus, for MeaningBERT, two sanity checks: identical sentences must score 100 and unrelated sentences 0. Baral (2026) observes that both checks are satisfied by any monotone function of lexical overlap, which is why they do not catch the failure studied here.

Inference and its held-out suites. Natural language inference supplies the missing dimension. Large-scale corpora (Williams et al., 2018; Nie et al., 2020) and fact verification (Thorne et al., 2018; Schuster et al., 2021) provide three-way entailment, neutral and contradiction labels, while targeted suites test what the aggregate hides: Geiger et al. (2020) probes downward monotonicity, where a lexical substitution inside the scope of a negation reverses the direction of entailment, and NaN-NLI (Truong et al., 2022) probes negation scoping below the clause. An inference model says whether the second sentence follows from or denies the first and nothing about how much meaning the two share; a preservation metric answers the converse. Marelli et al. (2014) is, to our knowledge, the only resource annotating both over the same pairs, which makes it the only place the composition can be calibrated.

3 A Signed Scale

We define the score so that +100 is identical meaning, 0 is no relation and −100 is opposed meaning. The decisive property is that the negative half measures active contradiction rather than the absence of shared meaning: two sentences about different topics score 0, not −100, since to contradict each other they must first be about the same thing. The score therefore carries two quantities. The magnitude asks what the sentences talk about together and is exactly what existing metrics estimate; the polarity asks, once the subject is shared, whether they assert the same thing or its negation.

We compose the two as

$$s = \text{clip}_{[-100,100]}(m \times (1 - \alpha p_{\text{contra}})), \quad (1)$$

with $m \in [0, 100]$ from a magnitude head and $p_{\text{contra}} \in [0, 1]$ from a three-way polarity head. The form is a product and not a difference: an unrelated pair has $m \approx$

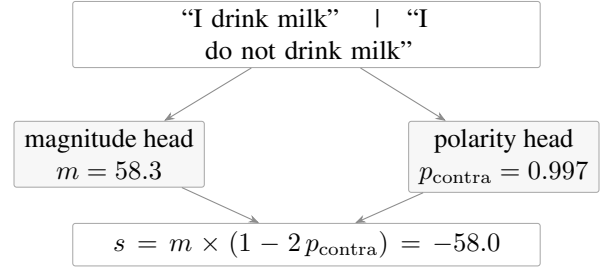


Figure 1: One pair, end to end. The two heads are read independently and Equation 1 turns a middling positive score into a negative one.

Pair	m	p_{contra} (%)	s
“I drink milk” / “I consume milk”	96.70	0.50	+95.80
“I drink milk” / “I do not drink milk”	58.30	99.70	−58.00
“I drink milk” / “the meeting is at three”	0.00	0.50	+0.00
“Two dogs are wrestling” / “there is no dog wrestling”	51.70	99.40	−51.10

Table 1: Measured, not illustrative. m from a released magnitude model, p_{contra} from our polarity head, s from Equation 1 at $\alpha = 2$. Rows two and four are the cases a magnitude scale gets backwards.

0 and scores near zero whatever its polarity, whereas a subtraction would let it drift negative and destroy the distinction the scale rests on. Figure 1 traces one pair through both heads, and Table 1 gives four measured cases.

The slope α governs how much contradiction is needed to flip the sign. A certain contradiction on a perfect magnitude lands at $100 \times (1 - \alpha)$, so the negative half is reachable only from $\alpha = 2$ upwards, and below it the scale is truncated. We treat reachability as a constraint on the search rather than as a term in the objective, since it defines the scale rather than trading against its other properties, and Section 7 reports what the constraint costs.

Two alternative designs were rejected. Typed penalties in the style of MQM, subtracting a cost per error category from a perfect score, give an interpretable and natively negative output but require an error taxonomy annotated over the same pairs, and they replace the target rather than extend it. Learning $[-100, 100]$ end to end requires manufacturing the target from categorical labels, which amounts to training a model on a conversion rule and then reporting the rule. Composition

Corpus	Rows	Labels	Use
VitaminC	488 904	FACT3	split
SICK	9 840	NLI3 + rel.	split
MoNLI	1 202	ent./neutral	held-out
NaN-NLI	258	NLI3	held-out

Table 2: Corpora. FACT3 is SUPPORTS, NOT ENOUGH INFO, REFUTES, mapped onto NLI3. *Split*: merged, then divided into train, development and test. SICK alone carries a relatedness score, which the composition is calibrated against.

avoids both problems, and it keeps each head separately falsifiable: when the composed score is wrong, the responsible half is identifiable.

4 Data

We merge four corpora carrying inference labels over sentence pairs (Table 2). Schuster et al. (2021) supplies the bulk, and its pairs are contrastive by construction, since a Wikipedia sentence was edited so that the claim its previous revision supported is refuted by the next. SICK carries both a relatedness score and an inference label over the same pairs, which makes it the only corpus the composition can be calibrated on. MoNLI and NaN-NLI never enter training and serve as held-out evaluation sets. PAWS (Zhang et al., 2019) was considered and rejected: its binary paraphrase label is not a polarity judgement, since two sentences that are not paraphrases need not contradict each other, and mapping *not a paraphrase* onto contradiction would teach the head that difference amounts to opposition.

Derived-polarity augmentation. A second corpus condition adds generated pairs whose polarity is derived rather than copied, because polarity is not symmetric the way magnitude is. If A contradicts B then B contradicts A , so a mirrored contradiction remains one; but “a dog is running” entails “an animal is running” and the converse is false, so a mirrored entailment loses its label rather than carrying a wrong one. Three families enter in equal quotas of 25 000: mirrored contradictions, identical pairs labelled entailment, and low-overlap random pairs labelled neutral. The last is what teaches that unrelated is not opposed. Equal quotas are necessary, since mirroring alone would take the contradiction class from 50 641 to 101 282 against 51 274 entailments and bias the head towards the one class whose sign inverts the score.

Splits. Development and test splits are drawn stratified by class and by corpus, 4 000 rows per class. Without stratification by corpus, SICK falls to 8% of the test split because VitaminC is fifty times its size, and model selection is then driven by a split that is almost entirely VitaminC; stratification raises SICK to 34%. We then remove source-sentence leakage in both directions, which matters because VitaminC reuses the same evidence across many claims and splits by case rather than by evidence: 5.6% of the test groups shared a source sentence with training. Appendix A gives the construction and the exact counts.

5 Experimental Setup

We train a three-way classification head on seven encoders $\times$ ten seeds (42–51) $\times$ two corpus conditions, for 140 runs. The conditions are RAW (152 060 training rows) and AUG (227 060). Encoders span three families, BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019) and DeBERTaV3 (He et al., 2023), and two sizes, and include both plain and inference-pretrained variants of the same architecture so that the effect of inference pretraining is measured at constant size. Checkpoints already fine-tuned for three-way inference keep their head when its class order matches ours and have their classifier rows permuted when the published order is reversed. All runs share one set of optimisation settings (Appendix B) on the hardware of Appendix C. The two halves of the metric cost very different amounts: the grid is roughly 500 GPU-hours, while the composition of Section 7 trains nothing, fits one parameter and runs on CPU in under half an hour.

We report macro- F_1 on the stratified test split, the AUC of $1 - p_{\text{contra}}$ ranking entailment above contradiction, accuracy on three generated sanity suites built from test-split sentences, and accuracy on the two held-out sets. Significance uses Welch’s t -test (Welch, 1947) between conditions and a paired test by seed between architectures, with Cohen’s d beside every p -value, since ten seeds with standard deviations near two tenths of a point make a difference of one tenth significant and meaningless at once. Holm-Bonferroni (Holm, 1979) across the fourteen condition tests changes no verdict, and we report uncorrected p -values.

6 Results

The polarity head is learnable, and not by surface statistics. The best configuration reaches $91.1 \pm 0.14\%$ macro- F_1 on a class-balanced test set of 12 000 pairs (Table 3), with a seed standard deviation of 0.14 points over ten runs. Table 6 places three references beside it. The majority class sits at the arithmetic floor of

Encoder	Condition	macro-F ₁	Sanity suites	NaN-NLI	MoNLI
DeBERTa-v3-large _{NLI}	RAW	91.10 ± 0.14	90.35 ± 0.47	72.95 ± 0.65	34.93 ± 3.13
	AUG	90.98 ± 0.31	98.76 ± 0.04	71.90 ± 1.83	31.23 ± 5.11
DeBERTa-v3-large	RAW	90.91 ± 0.10	88.89 ± 0.42	70.62 ± 1.17	22.79 ± 5.22
	AUG	90.70 ± 0.43	98.70 ± 0.40	69.11 ± 1.24	15.97 ± 5.84
RoBERTa-large _{MNLI}	RAW	89.52 ± 0.10	84.59 ± 1.07	68.06 ± 1.02	7.02 ± 1.10
	AUG	89.70 ± 0.10	98.69 ± 0.07	66.59 ± 1.09	8.34 ± 1.68
DeBERTa-v3-base _{NLI}	RAW	89.48 ± 0.15	87.46 ± 0.37	68.57 ± 0.62	11.47 ± 2.39
	AUG	89.56 ± 0.17	98.74 ± 0.08	67.40 ± 0.67	11.62 ± 2.05
DeBERTa-v3-base	RAW	88.81 ± 0.63	86.32 ± 1.74	66.40 ± 1.68	13.42 ± 2.96
	AUG	88.73 ± 0.35	98.60 ± 0.13	64.11 ± 1.92	10.87 ± 3.27
RoBERTa-base _{STS-B}	RAW	85.91 ± 0.15	78.67 ± 0.34	63.76 ± 0.99	4.60 ± 0.60
	AUG	86.22 ± 0.13	98.65 ± 0.08	62.60 ± 0.74	4.94 ± 0.64
BERT-base	RAW	77.72 ± 0.38	79.38 ± 0.65	46.36 ± 2.70	13.64 ± 5.52
	AUG	78.45 ± 0.26	98.31 ± 0.10	49.03 ± 1.13	17.48 ± 2.57

Table 3: Polarity head, mean and standard deviation over ten seeds, in percent. AUG adds the derived-polarity augmentation to RAW. NaN-NLI and MoNLI are held-out sets. **Bold**: best per column.

16.7%, Jaccard overlap between the two token sets cut by two thresholds fitted on development data reaches 39.4%, and a TF-IDF bag of unigrams and bigrams with a linear model reaches 45.6%. Between the floor and the fine-tuned head, token overlap covers 30.5% of the achievable range and TF-IDF 38.8%, so roughly two fifths of the task is surface statistics of the corpus and the head is worth 45 points above the best of the three.

Augmentation buys sanity, not performance. On the four DeBERTa encoders the macro-F₁ difference between conditions lies between -0.21 and $+0.08$ points and is not significant (p from 0.17 to 0.74); on the other three it is significant but below one point. The sanity suites move in a different regime (Table 4, Figure 2): $+8.4$ to $+20.0$ points, $p < 10^{-4}$ on all seven encoders, with d from 9.9 to 81.1. The effect sizes settle the matter, since $d = 81$ on RoBERTa-base_{STS-B} places the shift two orders of magnitude above the seed-to-seed noise, whereas the largest task effect anywhere in the grid is $d = 2.2$ for a difference of 0.73 points. Every encoder converges to 98.3–98.8% regardless of where it started, and the clearest cases are the weakest models: RoBERTa-base_{STS-B} moves from 78.7% to 98.7% and BERT-base from 79.4% to 98.3%. The augmentation does not make the head better at the task; it repairs how the head behaves at the ends of the scale, which is where a signed metric is read.

The sanity suites are built by the same three generators that produce the augmented training rows, over

different sentences but by the same construction, so an AUG model is evaluated partly on a distribution whose shape it has seen and the gain is not a claim of generalisation to arbitrary degenerate pairs. What the comparison establishes is that this behaviour is learnable and is not acquired incidentally: seven encoders trained without these rows converge nowhere near 98%.

The gain is concentrated on two families. Table 5 separates the three generated suites. Identical pairs are essentially solved before augmentation. Unrelated pairs rise from 92–96% to 99.3–99.7%: before augmentation, between 2.4% and 7.9% of pairs with nothing in common are predicted contradiction, which on the signed scale returns a negative score where the truth is zero. Mirrored contradictions rise from 74–75% to 96%, and on BERT-base from 50%, so the symmetry of contradiction is not a property these encoders hold on their own.

Inference pretraining buys reasoning under negation. Paired by seed, DeBERTa-v3-large_{NLI} beats its plain counterpart by $+0.19$ points of macro-F₁ ($p = 0.002$, $d = 1.39$) and by $+12.1$ points on MoNLI ($p < 10^{-4}$, $d = 2.58$). At base size the pattern inverts: the task gain is larger ($+0.67$ points, $p = 0.005$) and the MoNLI difference is not significant. Inference pretraining therefore supplies a capability the aggregate test set does not exercise, which is the argument for reporting held-out suites alongside it. MoNLI accuracy is low for every configuration, and the confusion

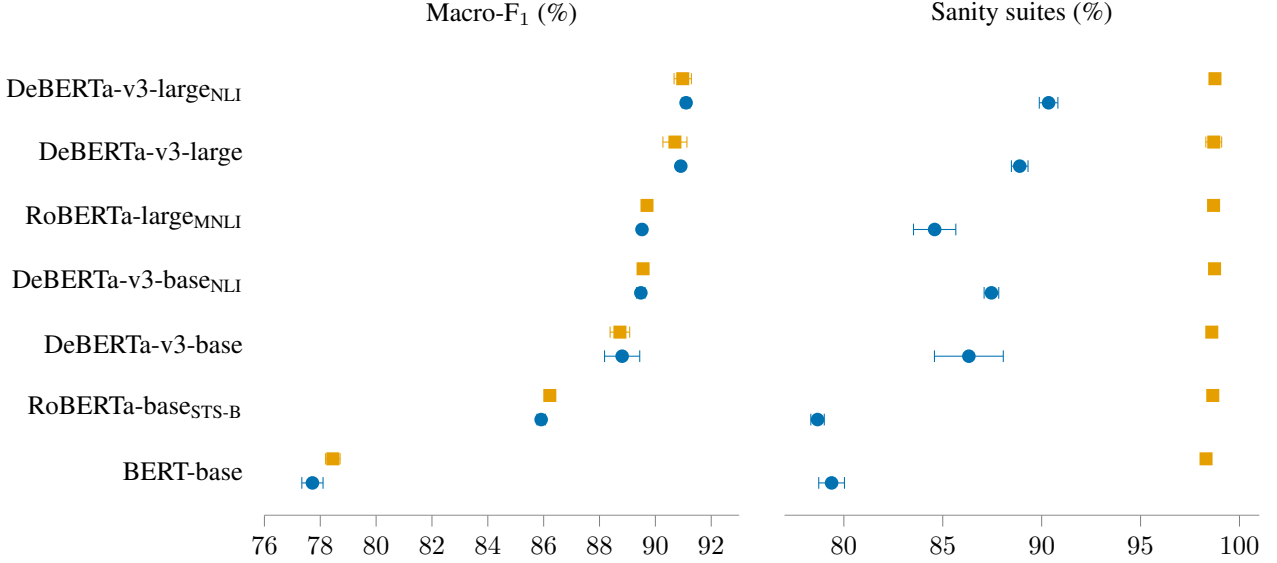


Figure 2: Augmentation moves one axis and not the other. **RAW** and **AUG**: means over ten seeds, bars one standard deviation. The panels do not share an x range.

Encoder	macro-F ₁		Sanity suites	
	Δ (pp)	d	Δ (pp)	d
DeBERTa-v3-large _{NLI}	$-0.11 \pm 0.11^\dagger$	-0.47	$+8.40 \pm 0.15$	$+25.43$
DeBERTa-v3-large	$-0.21 \pm 0.14^\dagger$	-0.67	$+9.81 \pm 0.18$	$+24.07$
RoBERTa-large _{MNLI}	$+0.19 \pm 0.05$	+1.79	$+14.10 \pm 0.34$	$+18.57$
DeBERTa-v3-base _{NLI}	$+0.08 \pm 0.07^\dagger$	+0.49	$+11.28 \pm 0.12$	$+42.24$
DeBERTa-v3-base	$-0.08 \pm 0.23^\dagger$	-0.15	$+12.28 \pm 0.55$	$+9.94$
RoBERTa-base _{STS-B}	$+0.31 \pm 0.06$	+2.22	$+19.98 \pm 0.11$	$+81.07$
BERT-base	$+0.73 \pm 0.15$	+2.22	$+18.94 \pm 0.21$	$+40.87$

Table 4: Effect of augmentation in points, AUG minus RAW, with the standard error of the difference and Cohen’s d (Welch, ten seeds). $\dagger$: not significant at $p < 0.05$. Every sanity difference has $p < 10^{-4}$.

matrix locates the failure precisely: the heads almost never call these pairs contradictions, with 0.2% of true entailments predicted contradiction on DeBERTa-v3-large_{NLI}, but 93% of true entailments are predicted neutral. They have not failed to notice the negation; they have failed to reverse the direction of entailment under it, applying the unnegated rule instead.

The headline transfers across corpora. Stratification leaves the test split one third SICK and two thirds VitaminC, two corpora of different provenance. Scored separately, DeBERTa-v3-large_{NLI} reaches 91.6% macro-F₁ on SICK against 90.4% on VitaminC, and 99.3% against 98.4% AUC. These four figures and the AUC of Table 6 come from a single run of the best configuration, since the grid stores predicted classes rather than the probabilities an AUC needs, and they carry no seed variance. The aggregate is therefore not an artefact of the larger corpus.

Polarity does not follow from preservation data.

Table 6 scores three released magnitude models on both tasks at once. The two columns do not order the models the same way: the weakest magnitude model, at $r = 0.323$, ranks polarity better than one with twice its correlation, and that second model, a competent magnitude estimator at $r = 0.640$, sits near chance at 53.7%. Nothing in a magnitude metric’s training signal distinguishes a contradiction from a failed paraphrase, since both receive the same low human rating, so no quantity of meaning-preservation data can produce polarity. The exception is instructive: DeBERTa-v3-large reaches 81.5% without ever seeing a polarity label, while a BERT-base model trained on the same preservation corpus reaches 53.7%, which places the separation in the encoder’s pretraining rather than in the preservation data.

An off-the-shelf inference model used as published ranks entailment above contradiction at 96.6%, above

Encoder	Condition	Identical	Unrelated	Mirrored
DeBERTa-v3-large _{NLI}	RAW	100.00 $\pm$ 0.00	95.91 $\pm$ 0.78	75.14 $\pm$ 1.20
	AUG	100.00 $\pm$ 0.00	99.73 $\pm$ 0.07	96.54 $\pm$ 0.13
DeBERTa-v3-large	RAW	100.00 $\pm$ 0.00	92.75 $\pm$ 0.60	73.93 $\pm$ 1.46
	AUG	99.99 $\pm$ 0.04	99.71 $\pm$ 0.13	96.41 $\pm$ 1.14
BERT-base	RAW	96.46 $\pm$ 0.16	91.99 $\pm$ 0.67	49.68 $\pm$ 1.74
	AUG	100.00 $\pm$ 0.00	99.33 $\pm$ 0.09	95.61 $\pm$ 0.29

Table 5: The three generated suites separately, ten seeds, in percent. Identical pairs must be entailment, unrelated pairs neutral, mirrored pairs contradiction.

Model	Preserv. r	Polarity AUC
<i>Magnitude metrics</i>		
MeaningBERT (published)	0.323	60.80
BERT-base, retrained	0.640	53.70
DeBERTa-v3-large, retrained	0.705	81.50
<i>Polarity heads, published then ours</i>		
DeBERTa-v3-large _{MNLI}	–	96.60
DeBERTa-v3-large _{NLI}	–	98.72
<i>Surface baselines</i>		
Token overlap, Jaccard	–	63.50
TF-IDF, linear model	–	68.50

Table 6: Both tasks on one test split. r is on the 1 536 human-annotated preservation pairs, excluding 116 generated controls that inflate it by up to +0.21; AUC ranks entailment above contradiction on our 12 000 pairs. Dashes: no preservation score. Three-way macro- F_1 of the baselines, 16.67, 39.36 and 45.64, against 91.10 for the fine-tuned head.

every magnitude model, which bounds what fine-tuning contributes. Against the grid mean, fine-tuning adds +22.7 points of macro- F_1 , and on the single run where both are measured it adds +2.1 points of AUC, so it improves both the ordering of p_{contra} and its calibration into three classes, but the ordering was largely available without it. A practitioner who needs only the sign may reach most of the way with no training at all.

7 Composing the Two Heads

We fit α on the SICK half of the development split and report on the SICK half of the test split, 4 116 pairs of which 712 are contradictions. The components are a released magnitude model and our best polarity head, DeBERTa-v3-large_{NLI}. Since no signed gold score exists, α is selected against three requirements multiplied together, so that sacrificing one cannot hide behind the others: contradictions must fall below zero, entailments must stay above it, and on the pairs whose

Slope	Contra. < 0	Entail. > 0	Rel. r	Floor
$\alpha = 1.00$	0.0	100.0	0.833	0
$\alpha = 1.25$	81.2	100.0	0.825	–25
$\alpha = 1.50$	82.7	100.0	0.816	–50
$\alpha = 1.75$	83.3	100.0	0.806	–75
$\alpha = 2.00$	83.7	100.0	0.795	–100
$\alpha = 3.00$	84.7	99.9	0.745	–100
Magnitude alone	0.0	100.0	0.852	0

Table 7: Composition on the SICK test half, in percent. The slope is fitted on development data, which selects $\alpha = 2$ (**bold**). Floor: the most negative score the scale can reach.

sign does not flip the score must still track human relatedness. A composition sending every contradiction to –100 and everything else to noise satisfies the first two and is excluded by the third. The development split selects $\alpha = 2$, the value the reachability constraint of Section 3 also forces.

Table 7 gives the result and Figure 3 shows it. The magnitude alone puts no contradiction below the line, which is a property of a non-negative scale rather than a weakness of that model. Composition puts 83.7% of them where they belong while retaining $r = 0.795$ with human relatedness on the unflipped pairs, against 0.852 for the magnitude alone. The cost of requiring the full scale is visible and small: $\alpha = 1.5$ scores 1.5% higher on the combined objective and truncates the negative half at –50.

The slope survives a change of magnitude model. Repeating the fit on the development split with two released checkpoints whose preservation correlations are 0.323 and 0.705, both select $\alpha = 2$ under the reachability constraint and $\alpha = 1.5$ without it. Both are variants of the same metric, so this tests the slope against model quality rather than against a structurally different magnitude estimator. On that split both also send the same 84.6% of contradictions below zero, which follows

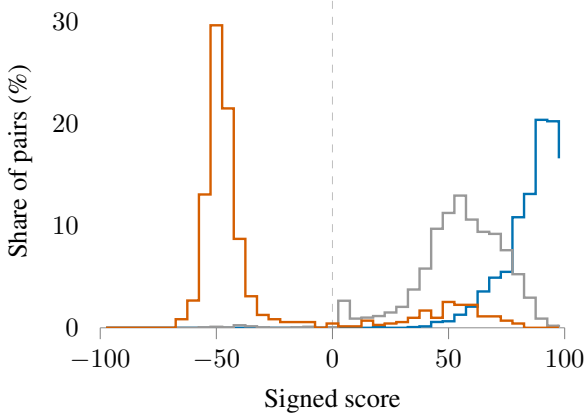


Figure 3: Signed score by gold class on the SICK test half at $\alpha = 2$, bins of width 5: **entailment** ($n = 1404$), **neutral** ($n = 2000$), **contradiction** ($n = 712$). The dashed vertical line marks zero, where the sign changes. SICK neutral pairs are related captions, not unrelated sentences, so they belong in the positive half.

Polarity head	Published	Fine-tuned
Contradictions below zero	97.61	83.71
Relatedness r	0.599	0.795
Combined objective	0.574	0.665

Table 8: One magnitude model composed with two polarity heads at $\alpha = 2$, SICK test half. First row in percent of the 712 contradictions. Higher is better throughout.

from Equation 1: with $m \geq 0$, the sign of the product is decided by p_{contra} alone and the magnitude sets only the amplitude.

What fine-tuning buys the composition. Composing the same magnitude model with the off-the-shelf head instead of ours inverts the trade-off (Table 8). The off-the-shelf head is the better sign detector, putting 97.6% of contradictions below zero against 83.7%, but it costs 0.196 of correlation with human relatedness and loses the combined objective by 14% relative. Figure 4 shows why: among the pairs to which it assigns a contradiction probability near 0.5, fewer than one in ten is a contradiction, so the product drags down the magnitude of pairs that are merely unrelated. The third requirement is what makes this visible, and it is the reason α is selected against a product rather than against the sign criterion alone.

The scale on simplification output. SICK and VitaminC are inference corpora; the metric is meant for the output of a simplification system. Scoring the 407

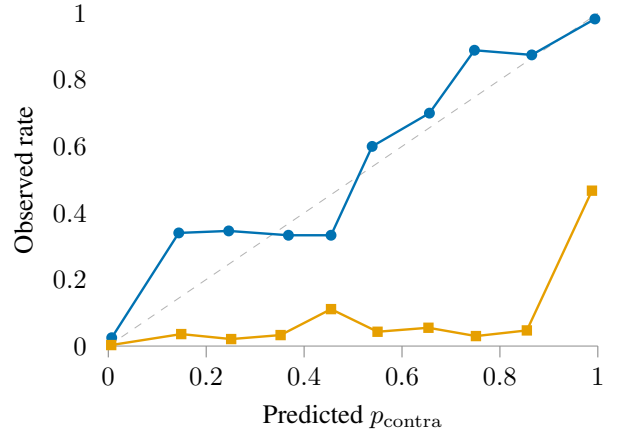


Figure 4: Reliability of p_{contra} on the SICK test half, ten bins. The dashed diagonal is perfect calibration, where a predicted probability equals the observed rate. The **fine-tuned head** tracks it, the **off-the-shelf head** falls far below.

human-annotated pairs of the preservation test set with the composed scale sends 4.2% of them below zero, and those pairs carry a mean human rating of 38.3 against 71.0 for the rest. The sign selects the simplifications humans rated worst. On a corpus that contains no deliberate contradictions, the sign costs 0.06 of correlation with the human rating, 0.637 against 0.697 for the magnitude alone.

8 Conclusion and Future Works

A meaning-preservation metric that cannot distinguish a paraphrase from a negation is not imprecise but confidently wrong on the pairs where precision matters, and the metric deployed today has exactly this property. Making the scale signed does not require a better magnitude model; it requires a second head and an explicit rule for combining them. The combination is cheap, training nothing beyond a three-way classifier and fitting one parameter, and it moves 84% of contradictions to the correct side of zero where the magnitude alone moves none. What it does not yet have is a human-annotated signed target against which the combining rule itself could be tested.

Limitations

While we believe the signed scale is a useful contribution to the evaluation of meaning preservation, we acknowledge six limitations that bound our claims and offer avenues for future research.

No signed gold standard. The composition is calibrated against three necessary conditions rather than

against human signed judgements, because none exist. A corpus of human-rated signed pairs would let the form of Equation 1 be tested rather than argued, and collecting one is the natural next step.

Augmentation costs monotonicity. The gain in sanity is paid for on MoNLI, which drops on three of seven encoders and on both large DeBERTa variants, by -6.8 points on the plain one and -3.7 on the inference-pretrained one. Generated identical and unrelated pairs teach the easy ends of the scale, not the reversal of entailment under negation.

The calibration corpus is thin on one side. SICK supplies only 495 development rows, of which 229 shared source sentences with its own test portion and were removed, leaving the development split at 2.3% SICK against 34% in test. Model selection is therefore driven by VitaminC while the slope is fitted on SICK alone, and the two numbers are the first thing to watch should they diverge.

The sign costs correlation. On both human-rated quantities the composition correlates worse than the magnitude alone: 0.795 against 0.852 on SICK relatedness, and 0.637 against 0.697 on preservation ratings. This is what should happen, since neither rating scale is signed and a correct negative score can only move a pair away from a non-negative target, but it means no existing human judgement can establish that the signed scale is the better metric. Only a signed annotation could, which is the first limitation above.

The sign rests on one head. Because the composition is a product with $m \geq 0$, the sign of the score is decided entirely by p_{contra} . A polarity head that is wrong about a pair is not corrected by a good magnitude model, and the 16.3% of SICK contradictions that stay positive are exactly the pairs the head misses.

Scope. All experiments are English and sentence-level, and run in fp32 on Pascal GPUs (Appendix C); encoders requiring more recent hardware were not evaluated.

Ethical Considerations

A meaning-preservation metric is deployed to decide whether a rewritten sentence may be shown to a reader in place of the original. When such a metric reports that a negated sentence preserves the meaning of its source, as we measure released models doing, the error

reaches the reader rather than the engineer. The risk is concrete in the settings where simplification is most useful, where a layperson may rely on the rewritten text without access to the original (Beauchemin et al., 2023). A signed scale does not remove that risk, but it makes the failure visible to the system consuming the score instead of hiding it inside a positive number. Our corpora also inherit the biases of the Wikipedia revisions and caption corpora they are drawn from, and a polarity head trained on them carries those biases into whatever it gates (Bender et al., 2021).

Availability

The corpus builder, the loaders, the training and composition code, the 140 result files and the script generating every table and figure in this paper are available at <https://anonymous.4open.science/r/signed-meaning-scale> and will be released under an open licence at URL on acceptance. The merged corpus inherits the most restrictive licence among its sources and will be redistributed under CC BY-NC-SA 4.0. The loaders are released separately so the corpus can be rebuilt from the original sources.

Use of Large Language Models

This work utilized Large Language Models (specifically Claude) to assist with the linguistic refinement, typographical errors, and LaTeX formatting of the manuscript (table generation, figure improvements). The authors retained full control over the scientific content, data analysis, and conclusions.

Acknowledgements

Removed for double-blind review.

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A Corpus Construction and Leakage Control

Merging and harmonisation. The four corpora are merged into one table with a unified three-way polarity column. VitaminC carries FACT3 labels, which we map onto NLI3 as SUPPORTS to entailment, NOT ENOUGH INFO to neutral and REFUTES to contradiction. SICK

is assembled from two public halves, one carrying the inference label and one the relatedness score; the join is verified to lose no more than 1% of rows, and in practice loses none.

Stratification. Development and test splits take 4 000 rows per class, drawn in equal share from each corpus by water-filling: a corpus smaller than its share is taken whole and the shortfall is redistributed over the larger ones. Without it, SICK falls to 8% of the test split and 0.8% of the development split.

Leakage. We group rows by source sentence and move whole groups, never rows. VitaminC reuses one piece of evidence across many claims, so 1 662 source sentences, 5.6% of the test groups, were also present in training. We drop 2 379 training rows and 229 development rows, and assert that train/dev, train/test and dev/test share no pair and no source sentence. The assertion runs again after augmentation, since a generated pair inherits its source sentences.

Augmentation quotas. The three generated families enter in equal quotas of 25 000, which takes the training set from 152 060 rows to 227 060. Mirroring alone would take the contradiction class from 50 641 to 101 282 against 51 274 entailments.

B Training Details

Optimisation. Table 9 gives the settings, which are identical for all 140 runs. Nothing is tuned per encoder: one setting across the grid is what makes the seven encoders comparable, and the cost is that no encoder sits at its own optimum.

Setting	Value
Objective	cross-entropy, three classes
Optimiser	AdamW, default β
Learning rate	10^{-5} , linear decay
Epochs	2
Effective batch	32
Micro-batch	8 (base), 4 (large)
Gradient accumulation	4 (base), 8 (large)
Max sequence length	256 tokens
Precision	fp32
Seeds	[42, . . . , 51]

Table 9: Optimisation settings, identical across the grid. The micro-batch is sized for the smallest card, the accumulation holds the effective batch at 32.

Head reuse. Four of the seven checkpoints ship a three-way inference head. We keep it when its class order matches ours and permute its rows when the published order is reversed, as it is for `roberta-large-mnli`. The alternative, training from a permuted head, is silent and costs an epoch of relearning the permutation.

Seeds. Seeds [42, . . . , 51] control initialisation of the new head, data order and dropout. Unless stated otherwise, every number reported in this paper is a mean over the ten with its standard deviation.

C Hardware and Compute Budget

Hardware. The grid runs on three NVIDIA Pascal-generation cards: a GTX 1080 Ti (11 GB), a Quadro P5000 (16 GB) and a Titan Xp (12 GB). All training is fp32, since the generation has no usable bfloat16. Two of the seven encoders wedge the GTX 1080 Ti reproducibly, which we handle by excluding those two from that card rather than by diagnosing the driver fault.

Budget. The 140 runs take roughly 500 GPU-hours in total. A base-encoder run takes 68 minutes on the RAW condition and 109 minutes on AUG; a large-encoder run takes 318 and 489 minutes. The composition of Section 7 trains nothing: it is two forward passes over 4 116 pairs plus a grid search over one scalar, and runs on CPU in under half an hour. No private API was used, and the total monetary cost is the electricity of hardware we already own.